

Literature Review: Efficient Anomaly Detection Architectures for Resource-Constrained Edge SOC's

Rejeesh Koshy

1 Introduction

The rapid expansion of decentralized networks has necessitated the shift from cloud-centric Security Operations Centers (SOCs) to Edge-native security architectures. However, deploying robust intrusion detection systems (IDS) on legacy or resource-constrained hardware (e.g., IoT gateways, older desktops) presents significant challenges. This review examines the shift towards algorithmic efficiency over model scale.

2 The Shift to Efficient AI Architectures

Recent advancements in Large Language Models, specifically the **DeepSeek-V3** architecture, have challenged the prevailing assumption that "larger is better" [1]. By utilizing Mixture-of-Experts (MoE) architectures, researchers have demonstrated that activating a fraction of parameters per token can achieve state-of-the-art performance with significantly reduced computational overhead [1]. This paradigm shift—prioritizing inference efficiency over raw parameter count—provides the theoretical foundation for implementing lightweight detection engines on limited hardware (4GB RAM).

3 Deep Learning vs. Traditional ML at the Edge

While Deep Learning (DL) models such as CNNs and LSTMs offer high detection accuracy, they are often unsuitable for edge deployment due to memory bottlenecks. Research by **Al-Hasedi et al. (2025)** indicates that DL models frequently exceed the memory and latency budgets of IoT devices [2]. Their comparative analysis suggests that tree-based Machine Learning models, while lighter, often require careful tuning to match DL performance.

Furthermore, **Zhang et al. (2025)** highlight that the high inference latency of Deep Neural Networks (DNNs) can render them inapplicable for real-time threat blocking on edge devices [3]. This supports the argument for exploring hybrid statistical-ML approaches that avoid the heavy matrix multiplications inherent in Neural Networks.

4 Isolation Forest as an Optimal Candidate

In the domain of unsupervised anomaly detection, the **Isolation Forest (iForest)** algorithm has emerged as a superior candidate for resource-constrained environments. A systematic review by **Togbe et al.** confirms that iForest offers a markedly superior computational footprint compared to One-Class SVMs or Autoencoders [4]. Its ability to isolate anomalies based on path length rather than density calculations makes it particularly resilient to the "curse of dimensionality" while maintaining low CPU usage.

5 Conclusion

The literature suggests a clear gap: while cloud-based SOC leverage massive DL models, there is a lack of optimized, hybrid frameworks for Edge SOC. This project addresses that gap by implementing a comparative engine (Statistical vs. Isolation Forest) to validate whether "DeepSeek-inspired" efficiency principles can perform effectively in a home-lab cybersecurity context.

References

- [1] DeepSeek-AI, "DeepSeek-V3 Technical Report," *arXiv preprint arXiv:2412.19437*, Dec. 2024.
- [2] A. Al-Hasedi et al., "Intrusion Detection on Resource-Constrained IoT Devices with Hardware-Aware ML and DL," *IEEE Access*, vol. 12, pp. 150-165, 2025.
- [3] Y. Zhang et al., "Lightweight and Accurate DNN-Based Anomaly Detection at Edge," *TU Delft Research Repository*, 2025.
- [4] M. Togbe et al., "Isolation Forest Based Anomaly Detection: A Systematic Literature Review," *IEEE Access*, 2022.